

Silvaneio Vieira dos Santos Junior

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Summary

My research focuses on Bayesian time series, state-space models, longitudinal data, and causal inference. I have worked on sequential inference for dynamic generalized linear models and related software for simulation-free Bayesian analysis, including the **kdGLM** package. Current projects include a state-space framework for surrogate evaluation with longitudinal outcomes and surrogate markers, with accompanying **OnlineSurr** R package published on CRAN, and sequential methods for time-series mixture models.

Education

The University of Texas at Austin

Ph.D. in Statistics & Data Science, in progress

Expected: 2029

Federal University of Rio de Janeiro (UFRJ)

M.Sc. in Statistics, 2024

Thesis: “Computational Methods for k -parametric Dynamic Generalized Linear Models”

Advisor: Mariane B. Alves

GPA: 4.0/4.0

Federal University of Rio de Janeiro (UFRJ)

B.Sc. in Statistics, 2022

GPA: 9.0/10.0 (Magna Cum Laude, 2nd in graduating class)

Technical Skills

Languages/Tools: R, Python, Stan, TensorFlow, Keras, Git, $\text{\LaTeX}$, SQL, Quarto/R Markdown.

Data/ML: state-space modeling, Bayesian inference, forecasting, GLMs/DGLMs, Markov chains, hierarchical models, numerical optimization, MCMC/Monte Carlo.

Software

kdGLM — R package for k -parametric dynamic GLMs (state-space time series).

Repository: github.com/silvaneiojunior/kdGLM.

OnlineSurr — R package for surrogate Evaluation for jointly longitudinal outcome and surrogate.

Repository: github.com/silvaneiojunior/OnlineSurr.

Publications

Peer-Reviewed

- Alves, M. B., Migon, H. S., Menezes, A., Pinheiro, E., **dos Santos Jr., S.** (2025). Dynamic Econometric Models: A State-Space Formulation. *Journal of Forecasting*, 44(8), 2494–2508.
DOI: <https://doi.org/10.1002/for.70017>.
Impact (venue): Journal of Forecasting SJR: 0.701.

- Freitas, W. V. de, dos Santos Jr., S., Alves, M. B., Nouér, S. A. (2024). Impacto da intervenção no uso de antimicrobianos em unidade de terapia intensiva (UTI). The Brazilian Journal of Infectious Diseases, 28(Suppl. 3), 104440.
DOI: <https://doi.org/10.1016/j.bjid.2024.104440>.
Impact (venue): SJR: 0.644; Q2; H-index: 59.

Under Review & Preprints

- dos Santos Jr., S., Parast, L. (2026). A Causal Framework for Evaluating Jointly Longitudinal Outcomes and Surrogate Markers: A State-Space Approach Preprint: <https://arxiv.org/abs/2604.12882>.
Dissemination: arXiv first posted 2026-04-14.
- Alves, M. B., Migon, H. S., dos Santos Jr., S., Marotta, R. (2025). k-Parametric Dynamic Generalized Linear Models: A Sequential Approach via Information Geometry. *Submitted to Computational Statistics & Data Analysis*, status: under review (2025).
Preprint: <https://arxiv.org/abs/2201.05387>. *Impact (venue)*: CSDA SJR: 0.885; Q1; H-index: 138. *Dissemination*: arXiv first posted 2022-01-14.
- dos Santos Jr., S., Alves, M. B., Migon, H. S. (2024). kDGLM: a R package for Bayesian analysis of Generalized Dynamic Linear Models.
Preprint: <https://arxiv.org/abs/2403.13069>. *Dissemination*: arXiv first posted 2024-03-19.
- Barreto, D., Borges, R. B. de R., Costa, B., dos Santos Jr., S. (2023). New Weighting Strategies for WENO Schemes.
Preprint: <https://arxiv.org/abs/2311.09332>.

Research Experience

Research Assistant, Laboratory of Applied Mathematics (LabMA), UFRJ *Feb 2019–Apr 2022*
Project: “BR-EMS: Life table for the Brazilian insured population”.

Advisor: Bruno Costa.

Topics: Biometric table; Machine Learning; Anomaly detection; Clustering.

Tools: Python, TensorFlow/Keras, SQL, Google Cloud, Git.

Scientific Initiation (PIC CNPq)

Jun 2018–Jan 2019

Junior Scientific Initiation (PIC Jr. CNPq)

Jan 2015–Dec 2015

Grants, Fellowships & Scholarships

CAPES M.Sc. Scholarship (Brazilian federal agency)

May 2022–Apr 2024

Concurrent with LaMA/UFRJ research scholarship.

LaMA/UFRJ M.Sc. Research Scholarship

May 2022–Apr 2024

Concurrent with CAPES scholarship.

LaMA/UFRJ Undergraduate Research Scholarship.

Feb 2019–Apr 2022

Awards & Honors

Brazilian Public-School Math Olympiad (OBMEP): Bronze (2014, 2015); Honorable Mention (2012, 2013).

Talks & Presentations

- **A causal framework for evaluating jointly longitudinal outcomes and surrogate markers: A State-Space Approach.** S. dos Santos Jr. and L. Parast. 26^o National Symposium of Probability and Statistics (SINAPE 2026), oral session, Gramado, RS, Brazil, September 13–18, 2026.
- **A causal framework for evaluating jointly longitudinal outcomes and surrogate markers: A State-Space Approach** (invited). S. dos Santos Jr. and L. Parast. 9th International Workshop in Sequential Methodologies (IWSM 2026), invited session, American University, Washington, DC, June 1–4, 2026.
- **Modeling of Varicella Hospitalization Counts Via Dynamic Poisson Regressions.** *Authors:* Matheus Raposo, Silvano dos Santos Jr.; *Supervisor:* Mariane B. Alves. XLIV Giulio Massarani Journey of Scientific Initiation (Jornada Giulio Massarani de Iniciação Científica), Federal University of Rio de Janeiro (UFRJ), Rio de Janeiro, Brazil, May 29–June 2, 2023. **Honorable Mention (section prize).**
- **Neural Networks and the Numeric Resolution of Differential Equations.** *Authors:* Daniel Barreto, Silvano dos Santos Jr.; *Supervisor:* Bruno Costa. XLIII Giulio Massarani Journey of Scientific Initiation, UFRJ, Rio de Janeiro, Brazil, March 14–18, 2022. **Honorable Mention (section prize).**
- **Anomaly Detection in Biometric Tables through Long Short-Term Memory Network.** *Author:* Silvano dos Santos Jr.; *Supervisor:* Bruno Costa. XLII Giulio Massarani Journey of Scientific Initiation, UFRJ, Rio de Janeiro, Brazil, March 22–26, 2021. **Honorable Mention (section prize).**

Research Interests

Bayesian time series; state-space models; hierarchical models; nonparametric inference; causal inference.